

spread when the order was released to the market. This measure is usually expressed in basis points. That is:

$$Cost = Side \cdot \frac{P_{avg} - P_0}{P_0} \cdot 10^4 bp$$

where,

P_{avg} = average execution price of the order

P_0 = arrival price

$$Side = \begin{cases} +1 & \text{if Buy} \\ -1 & \text{if Sell} \end{cases}$$

Pre-trade *of* pre-trades. Another technique that has become more popular and is being used by portfolio managers is known as the pre-trade of pre-trade approach. Investors use pre-trade estimates provided by multiple broker-dealers and/or vendors for various order sizes, strategies, and stocks. These data points are used as input to the I-Star model, the results of which form a general consensus model of the industry. The technique is further described in Chapter 5, Estimating I-Star Model Parameters, as well as “Creating Dynamic Pre-trade Models: Beyond the Black Box” (Journal of Trading, Fall 2011).

Imbalance size issues. Each of the previously discussed methodologies to derive our order imbalance size is accompanied by some inherent limitations. These include:

1. *Misidentification.* Imbalance is inferred from the trade and may misidentify buys as sells and vice versa.
2. *Survivorship Bias.* Investors often allow orders that are trading well (inexpensive) to continue to trade and cancel those orders that are underperforming (expensive)
3. *Small Orders.* Large concentrations of small orders cause results to be skewed to be more accurate for small trades and potentially less accurate for large trades.
4. *Incomplete Data Set.* B/Ds and vendors are often not familiar with investors and portfolio managers’ intentions. They often observe day orders from the fund only (the fund may give a large multi-day order to different brokers each day in order to disguise their trading intentions).
5. *Over-fitting.* The universe of trades is executed in a very similar manner, making it difficult to perform what-if analysis and evaluate alternative trading strategies.